

Homework**Polona Oblak**

Let V be a vector space with basis $\mathcal{B} = \{b_1, \dots, b_n\}$ and its dual basis $\mathcal{B}^* = \{\beta_1, \dots, \beta_n\}$. For $\alpha_1, \alpha_2 \in V^*$ let $T_{\alpha_1, \alpha_2}: V \times V \rightarrow \mathbb{R}$ be a function defined by

$$T_{\alpha_1, \alpha_2}(v_1, v_2) = \det \begin{bmatrix} \alpha_1(v_1) & \alpha_1(v_2) \\ \alpha_2(v_1) & \alpha_2(v_2) \end{bmatrix}.$$

- (1) Prove that for all $\alpha_1, \alpha_2 \in V^*$ function T_{α_1, α_2} is a $(0, 2)$ -tensor.
- (2) Compute $T_{\beta_k, \beta_\ell}(b_i, b_j)$ for $i, j, k, \ell \in \{1, 2, \dots, n\}$.
- (3) Prove that $T_{\beta_k, \beta_\ell} = \beta_k \otimes \beta_\ell - \beta_\ell \otimes \beta_k$ for $k, \ell \in \{1, 2, \dots, n\}$.

The solution of the problem needs to be submitted to the appropriate mailbox on Ucilnica until a day before the exam, but preferably in a week. The solutions should be written by hand (on paper and scanned or on a tablet) and should contain clear and well-explained proofs, procedures, explanations, etc. At the end, add a short statement confirming that the submitted solution is your own work. Also list the names of people you discussed the assignment with (if any) and clearly indicate which parts of the solution were developed by you and which parts were assisted by AI tools (if any).